LIE DETECTION FROM VIDEOS

TECHNICAL REPORT

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ABSTRACT

We propose a multimodal approach for lie detection from videos that leverages audio, visual, and linguistic cues extracted from natural conversations. Traditional deception detection methods often depend on invasive physiological measurements, which limit their practical application; our method avoids this limitation by analyzing non-invasive behavioral signals. We evaluate our framework on the DOLOS dataset Guo et al. [2023], the largest game show-based deception dataset with 1,675 video clips from 213 participants, demonstrating its effectiveness in real-world settings. Our model enhances parameter-efficient crossmodal learning by integrating uniform temporal adapters and a hierarchical temporal processing architecture designed for long-form videos. By fusing multimodal features and employing a composite loss function to optimize cross-modal representation, our approach outperforms existing methods.

The project code is available at `https://github.com/shashuat/deception-detection`.

The demo video is available at drive link.

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Contents

1 Introduction **3**

2 Background **3**

2.1 Lie Detection . . . 3
2.2 DOLOS Dataset . . . 3
2.3 Video Understanding . . . 3
2.4 Adapter Modules . . . 3

3 Methodology **4**

3.1 Baseline Approach . . . 4
3.2 Enhanced Multimodal Fusion . . . 4
3.3 Loss Function Implementation . . . 4
3.4 Hierarchical Temporal processing . . . 4

4 Experiments **5**

4.1 Implementation Details . . . 5
4.2 Observation . . . 5
4.3 Training Results . . . 5
4.4 Model analysis . . . 6

5 Future Work **6**

6 Conclusion **6**

A Appendix **8**

A.1 Original PECL Framework . . . 8
A.2 Architecture . . . 8
A.3 Chunk Processing . . . 9
A.4 Model Variants List . . . 9
A.5 CNN activation . . . 10
A.6 Modality overfitting exemple . . . 10
A.7 A demo sample Inference . . . 11

1 Introduction

Motivation Misinformation has become increasingly prevalent in public discourse, especially in political communications and the media. The ability to detect deception accurately and noninvasively represents a significant challenge. Traditional methods of lie detection have relied on the measurement of physiological signals such as electrocardiogram (ECG) readings or brain activity patterns. Although these approaches can be effective, they are inherently invasive, require specialized equipment, and cannot be deployed for real-time analysis in natural settings. But by analyzing subtle behavioral cues from facial expressions, body language, speech patterns, and linguistic content. AI systems can potentially identify deceptive behavior without physiological sensors. This capability would enable real-time deception detection in natural conversational contexts, representing a substantial advancement over existing methods.

Objectives Develop a model with fusion mechanisms to integrate audio, visual, and linguistic features for deception detection in long-form videos while having some model interpretability. The goal is to identify specific segments and cues associated with deceptive behavior.

2 Background

2.1 Lie Detection

Several approaches to enhance the accuracy of deception detection by leveraging audio-visual cues. For instance, [Krishnamurthy et al., 2018] developed a multimodal neural model that combines features from video, audio, and text, achieving high accuracy in detecting deception in real-life videos. [Abdelwahab et al., 2024] and [Touma et al., 2023] have explored the use of advanced neural networks and fusion techniques to integrate audio and visual data, such as Feature-wise Linear Modulation (FiLM) and Graph Convolutional Networks (GCNs), which have shown promising results in capturing nuanced interactions between modalities. Additionally, voting-based multimodal methods have been proposed, utilizing CNNs, SVMs, and Word2Vec to analyze images, audio, and text, respectively, and have demonstrated superior performance over state-of-the-art models. Recently, [Ji et al., 2024] introduce a method to leverage knowledge transfer from facial expression datasets to improve deception detection accuracy, highlighting the importance of integrating diverse modalities and features.

2.2 DOLOS Dataset

The DOLOS dataset [Guo et al., 2023] provides naturalistic deception data from gameshow environments, where participants are strongly motivated to deceive. Similarly, court datasets (such as [Pérez-Rosas et al., 2015]), which capture real-life testimony, offer authentic instances of deception driven by genuine intent. In contrast to laboratory settings, where the stakes of lying are lower, both DOLOS and court datasets reflect more realistic deceptive behavior. DOLOS is notable as the largest gameshow-based deception dataset, comprising 1,675 video clips from 213 participants engaged in strategic deception.

2.3 Video Understanding

[Bertasius et al., 2021] demonstrated that pure transformer architectures can be effectively applied to video analysis by employing separate spatial and temporal attention mechanisms. This method processes video frames using self-attention both within individual frames and across consecutive frames, analogous to our model's strategy for analyzing facial expressions over video segments. The TimeSformer model showed that decomposing spatial and temporal attention yields computational efficiency and robust motion pattern capture. For long-form video analysis, hierarchical approaches are particularly effective, as evidenced by SlowFast Networks [Feichtenhofer et al., 2019]. In a similar vein, our hierarchical temporal processing framework examines videos at both the segment and sequence levels, enabling the identification of deceptive cues that occur over varying temporal scales.

2.4 Adapter Modules

Parameter efficiency is essential for practical video analysis. [Pfeiffer et al., 2020] introduces AdapterFusion, which incorporates task-specific adapter modules into pre-trained transformers while adding only a minimal number of parameters. This method facilitates effective knowledge transfer and mitigates catastrophic forgetting. In our work, we leverage the ViT model for transfer learning and train only the adapter modules to preserve parameter efficiency.

3 Methodology

3.1 Baseline Approach

Our baseline methodology (see Figure 2) reimplements the Parameter-Efficient Crossmodal Learning (PECL) framework proposed by [Guo et al., 2023]. This approach integrates Uniform Temporal Adapters (UT-Adapters) within transformer-based encoders for both video and audio modalities. In our implementation, we employ Wav2Vec2 [Baevski et al., 2020] as the backbone for audio encoding and the Vision Transformer (ViT) [Dosovitskiy et al., 2020] for visual feature extraction, with modality-specific CNN modules used for input tokenization. The UT-Adapters, which consist of linear projection and 1D convolutional layers, are strategically inserted in parallel with Multi-Head Self-Attention (MHSA) and Multi-Layer Perceptron (MLP) modules to efficiently capture local temporal dynamics while preserving the transformer's global attention capabilities. For crossmodal integration, we adopt the Plug-in Audio-Visual Fusion (PAVF) module, which computes crossmodal correlation matrices to facilitate bidirectional attention between modalities, thereby enabling the model to identify crucial audio-visual correspondences for deception detection. By fine-tuning only the adapter parameters and keeping the pre-trained weights frozen, this parameter-efficient approach effectively mitigates overfitting and substantially reduces computational requirements compared to full model retraining, all while leveraging the robust knowledge acquired from large-scale audio and visual tasks.

3.2 Enhanced Multimodal Fusion

We extend 3 the baseline fusion strategy with several improvements drawn from recent advances in natural language processing and multimodal representation learning:

Linguistic Features To capture semantic cues and linguistic patterns tied to deceptive behavior, we incorporate Whisper Radford et al. [2022] for speech transcription and a BERT-based Devlin et al. [2019] module for text encoding. These components add rich linguistic features beyond raw audio or visual signals.

Cross-Attention Inspired by Transformer-based techniques Vaswani et al. [2017], we introduce a cross-attention mechanism that interlinks the latent spaces of different modalities. Rather than treating each modality independently, our cross-attention block allows the model to highlight relevant features from other modalities, creating more coherent joint representations.

Gated Attention To further balance the contribution of each modality, we implement a gated attention mechanism Dauphin et al. [2017]. This gating adaptively weights each modality's attended features, permitting the model to strengthen or diminish specific modalities depending on their relevance. While we began with a simple weighted-sum fusion, adopting learnable gating provides a more flexible and data-driven means of controlling each modality's influence.

3.3 Loss Function Implementation

We adopt a weighted loss that merges the primary deception detection objective, modality-based tasks, and (optionally) sub-label supervision. Specifically, the main detection term is a cross-entropy on fused representations, while a second cross-entropy averages over unimodal outputs, with both terms weighted by λ_1 and λ_2. If sub-labels are present, an additional cross-entropy term is included. In practice, sub-label supervision yielded limited gains. We also experimented with a cross-modal contrastive loss (inspired by CLIP) but found no improvement. All losses are computed via standard cross-entropy in PyTorch, with gradient clipping applied each batch to maintain stability.

$$\mathcal{L}_{\text{final}} = \alpha \left(\lambda_1 \, \mathcal{L}_{\text{main}} + \lambda_2 \mathcal{L}_{\text{modality}} \right) + \beta \mathcal{L}_{\text{sub_labels}}$$

3.4 Hierarchical Temporal processing

The hierarchical architecture 4 processes latent embeddings extracted from video segments in a two-tier approach. First, input videos are divided into sequential chunks, each represented by high-dimensional embeddings (3072 dimensions) that we get from the original model. These embeddings are then processed through a sequence learning pipeline ('HierarchicalDeceptionModel' class). The architecture begins by projecting the high-dimensional chunk embeddings into a lower-dimensional space (768 dimensions) and applies positional encoding to preserve sequential information. Then, transformer encoder with multiple layers (default is 4 layers with 8 attention heads) to capture temporal relationships between sequential chunks. This transformer architecture enables the model to learn both local and long-range dependencies across the video timeline. The model then makes predictions at two distinct levels:

chunk-level classifications through a linear classifier that determines deception probability for each individual chunk, and a sequence-level classification through a more complex classifier (using a bottleneck architecture with layer normalization and GELU activation) that aggregates information across the entire sequence via mean pooling to produce a final deception score. This dual-level approach allows the model to simultaneously identify deceptive segments while considering the overall context of the sequence.

4 Experiments

We investigated several variants of our approach by varying the fusion strategies and modality combinations. Table A.4 summarizes the configurations explored in our experiments.

4.1 Implementation Details

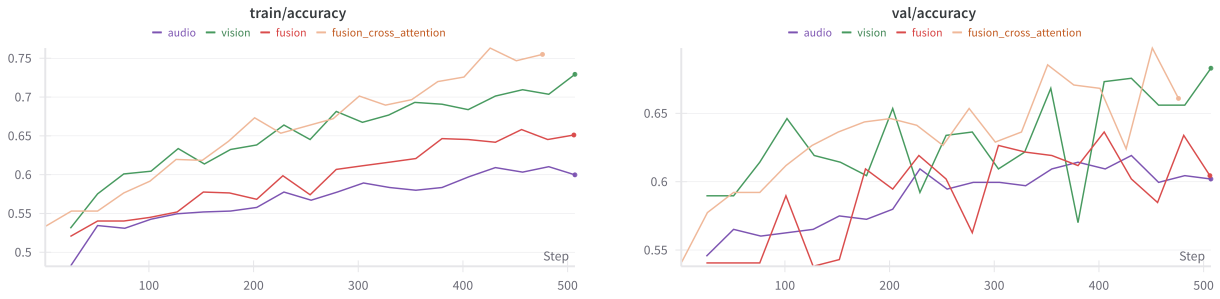
We implemented our multimodal deception detection framework using PyTorch. All models were trained on NVIDIA GPU with CUDA. For reproducibility, we fixed random seeds across all experiments (torch, numpy, and Python's random libraries). Our train configuration includes a learning rate of 1e-4, a batch size of 12, and a 20-epoch training schedule with an efficient convolution-based adapter. Optimization was performed using AdamW with a 1e-5 weight decay to mitigate overfitting. To stabilize training, we employed a two-phase learning rate schedule: a linear warm-up over the first epoch (from 10% to 100% of the base learning rate) followed by cosine annealing down to 1e-6. Our loss function was designed for multi-task learning, incorporating a weighted combination of main classification loss, auxiliary task losses (if enabled), and sub-label supervision, ensuring robust feature learning. Gradient clipping was applied to prevent instability due to exploding gradients, contributing to overall training stability and improved convergence.

4.2 Observation

Our preliminary experiments reveal that the MUMIN features alone can achieve an accuracy of approximately 65% when used with XGBoost, suggesting that they capture significant latent information. This observation indicates that these features possess inherent discriminative power, potentially serving as an effective feature selector for more complex deep learning models. The performance of the baseline fusion model, which incorporates these features, demonstrates that integrating multimodal signals can yield results comparable to those achieved by traditional machine learning techniques. This finding underscores the importance of further exploring and refining feature extraction and fusion methods to fully exploit the latent information present in the data.

4.3 Training Results

The final classification metrics show the model achieving about 70% overall accuracy and performing notably well on detecting deception, with higher recall for deception (81%) than for truth (56%). In practical terms, the model is more confident identifying deceptive statements than confirming truthfulness. This suggests the multimodal fusion effectively picks up on behavioral cues for deception, yet it can mislabel genuine statements as lies. Surpassing a naive 50% baseline on real, unscripted data indicates meaningful signal capture, though there remains a gap in improving truth-class performance and reducing potential false positives.



(a) Training Accuracy

(b) Validation Accuracy

Figure 1: Training and Validation Accuracies of our model (yellow).

Throughout training, accuracy fluctuates because the system must align multiple modalities (visual, audio, and text) while grappling with natural variation in the data. Our fusion strategy seems particularly capable of extracting cross-modal correlations, yet performance still varies depending on the modality. In some runs, the text branch dominates when dropout is not high enough (> 0.5), causing the gate to down-weight other inputs and overfit on text features, this leads to near-perfect training accuracy but around 50% on the validation set. Such behavior illustrates the challenge of balancing contributions from each modality. Additionally, the model is highly sensitive to initialization, and reinitializing can significantly shift outcomes. Addressing these issues, such as through more robust pretraining, remains an interesting subject for further exploration.

4.4 Model analysis

A qualitative analysis of the model's behavior offers additional insight into its decision-making process. The activation map of the first CNN layer (see Figure 5) reveals that the network emphasizes critical facial regions, including the eyes, nose, and eyebrows, aligning with the significance of the corresponding MUMIN features in deception detection. We also analyzed the cross-attention maps between modalities to understand the interplay among visual, audio, and textual cues. Although these maps did not show clearly defined patterns, they indicate that the model dynamically adjusts the weight of each modality during inference. However, we occasionally observed issues with the gating mechanism, which sometimes overemphasizes one modality over others, resulting in overfitting (see Figure 6). Further investigation into these intermodal relationships may uncover additional insights into how deceptive cues are integrated and guide the development of improved fusion strategies.

5 Future Work

3D Vision Transformer for Spatiotemporal Analysis. A key direction is to explore a 3D Vision Transformer (ViT) for capturing richer temporal and spatial correlations in video-based deception detection. The idea is to tokenize video frames into spatiotemporal patches and leverage self-attention to model motion cues and expression dynamics. Although our initial tests with a 3D ViT were computationally heavy and yielded limited performance gains, further optimization or more powerful hardware could make this approach more practical in the future.

Pretraining of the model We did not manage to tap into the full potential of the MUMIN labels during this project, but they still present a promising avenue for a dedicated pretraining phase. For instance, an approach such as SimCLR could cluster representations of truth versus lies prior to the main training. We plan to investigate this idea further in future work.

Refined Framework and Real-Time Processing. Substantial effort has gone into refactoring the existing codebase: addressing compatibility pitfalls, updating outdated libraries, and fixing architectural errors such as missing activation layers. Looking ahead, we intend to refine the framework further, with the eventual goal of integrating a dedicated graphical interface that supports real-time video processing.

6 Conclusion

We presented a multimodal system for lie detection that integrates visual, audio, and textual signals to capture subtle cues of deceptive behavior. By using efficient adapters and hierarchical temporal modeling, our approach simplifies fine-tuning while handling long-form videos more effectively than previous methods. Experimental results on the DOLOS dataset show that the proposed framework outperforms the baseline model, achieving around 70% accuracy and demonstrating especially strong recall for deceptive clips, which is fascinating as it also aligns with findings in human deception research: deception often produces more distinctive behavioral markers than truth-telling. Finally, Cross-modal correlations among facial expressions, voice patterns, and linguistic features are crucial for identifying lies in natural settings. Although some challenges persist with modality-specific overfitting and fluctuations in training stability, these findings confirm the value of rich multimodal fusion for deception detection and highlight promising avenues for further research into spatiotemporal transformers and more robust pretraining.

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A Appendix

A.1 Original PECL Framework

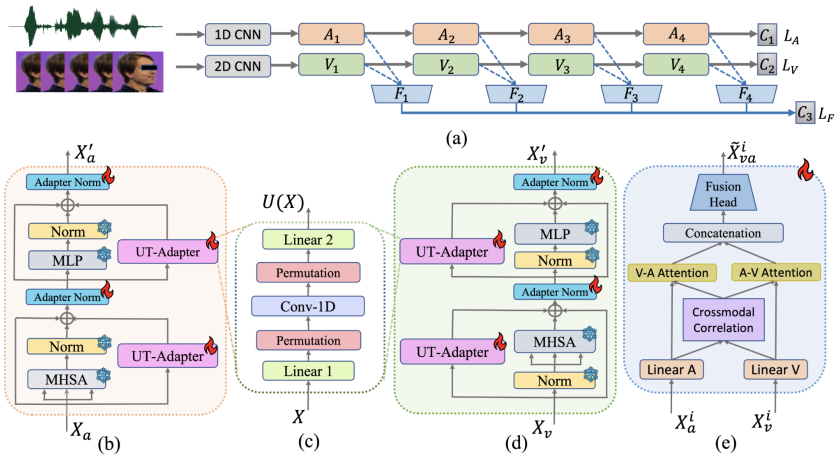


Figure 2: Original PECL Framework.

A.2 Architecture

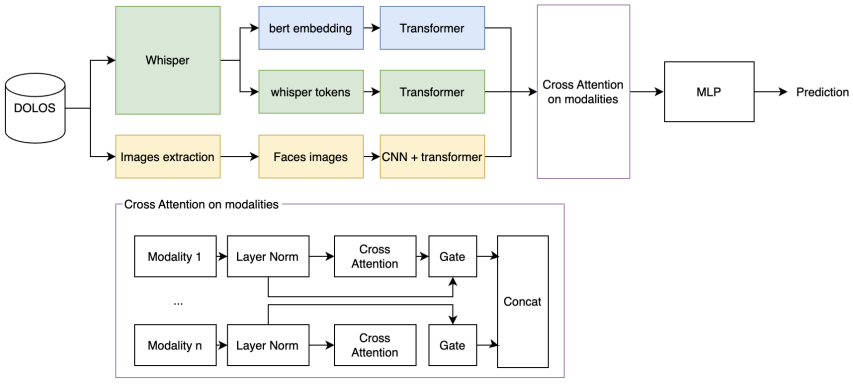


Figure 3: Our Model Architecture

A.3 Chunk Processing

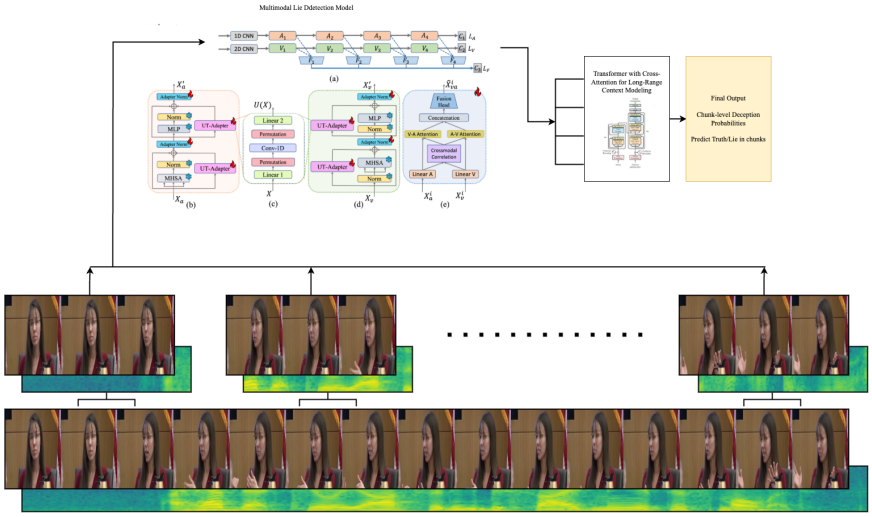


Figure 4: Chunk Processing for long videos.

A.4 Model Variants List

Fusion Strategies

- **concat**: Simple concatenation of modality embeddings.
- **cross2**: Cross-modal fusion using learnable correlation weights.
- **cross_attention**: Advanced cross-modal attention with multiple transformer layers.

Modality Combinations

- **Visual-only**: Using face encoder.
- **Audio-only**: Using Wav2Vec2 encoder.
- **Visual + Audio**: Combining both modalities.
- **Visual + Whisper + Audio + Text**: Adding linguistic features from transcripts.
- **Full multimodal**: Faces + Text + Whisper tokens.

Adapter Configurations

- **nlp**: NLP-style adapters with bottleneck layers.
- **efficient_conv**: Convolutional adapters with spatial processing.

A.5 CNN activation

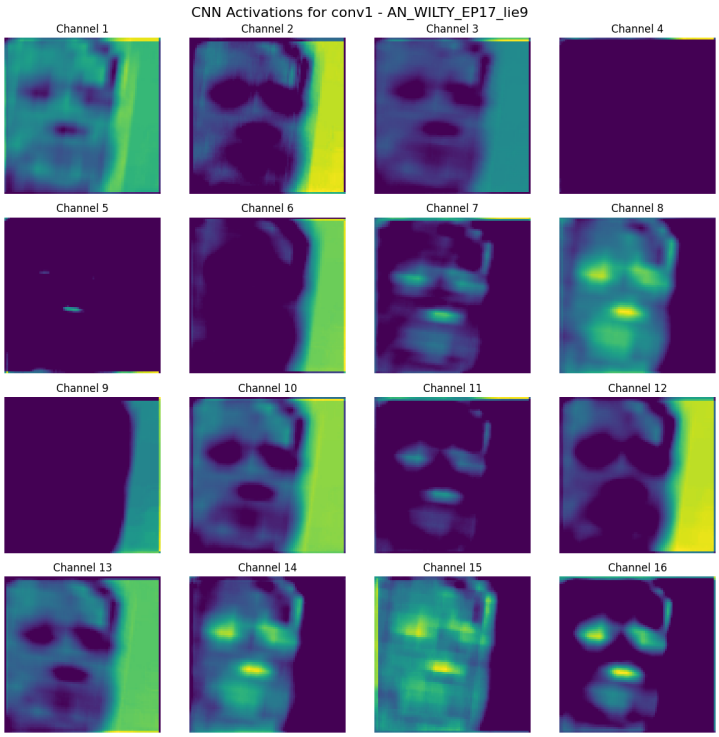
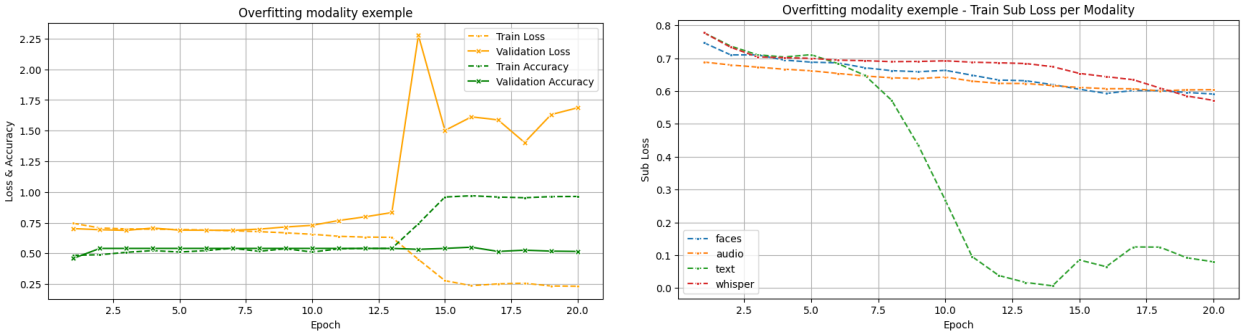


Figure 5: CNN Layer 1 Activation

A.6 Modality overfitting exemple



(a) Total Loss & Accuracy evolution

(b) Train Loss over Modalities

Figure 6: Exemple of overfitting model link to the gate mechanism.

A.7 A demo sample Inference

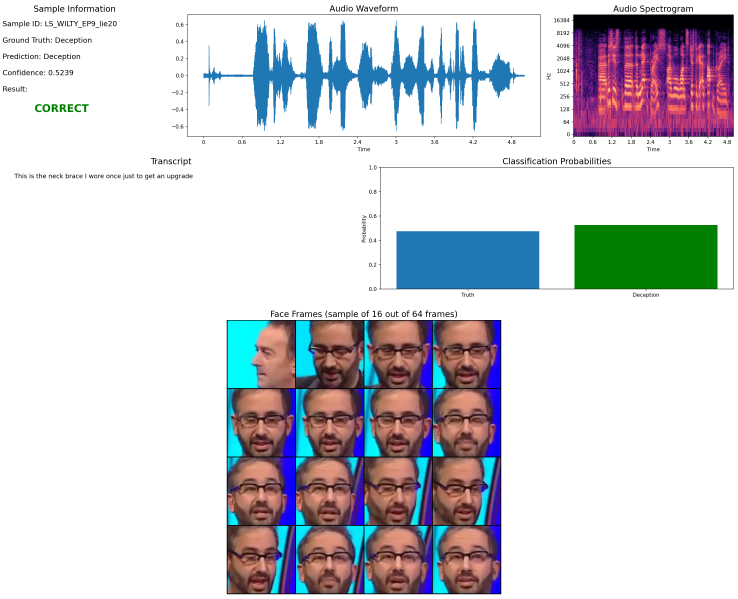


Figure 7: Processing a demo sample.